

Problem

A three-phase, four-wire system operating with a 480-V line voltage is shown in Fig. 12.71. The source voltages are balanced. The power absorbed by the resistive wye-connected load is measured by the three-wattmeter method. Calculate:

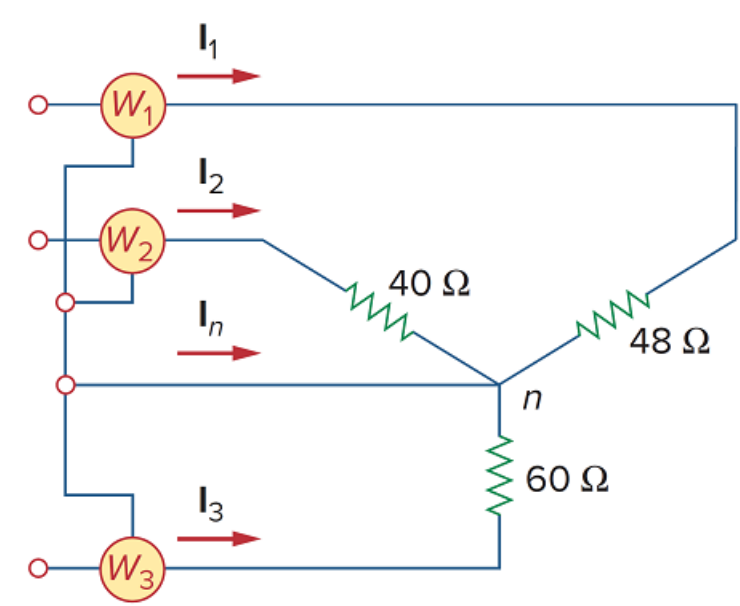
(a) the voltage to neutral

(b) the currents I_1 , I_2 , I_3 , and I_n

(c) the readings of the wattmeters

(d) the total power absorbed by the load

Figure 12.71

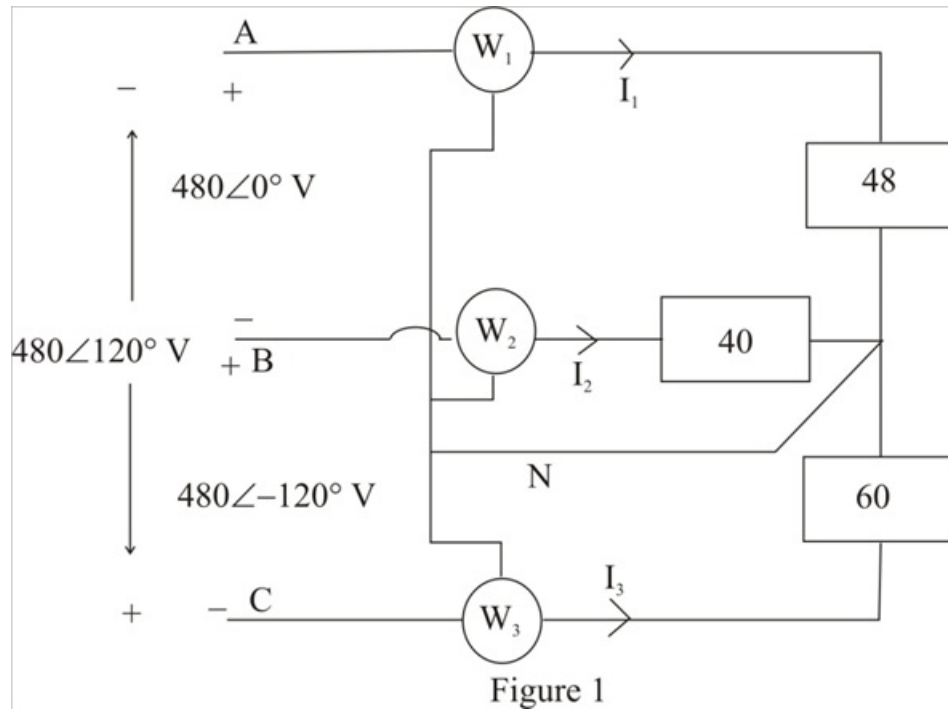


Step-by-step solution

Step 1 of 12

Refer to Figure 12.71 in the text book.

Consider the circuit diagram shown in Figure 1.



Step 2 of 12

(a)

Calculate the value of the voltage V_{AN} .

$$V_{AN} = \frac{V_{AB} \angle -30^\circ}{\sqrt{3}}$$

Substitute $480 \angle 0^\circ$ V for V_{AB} .

$$\begin{aligned} V_{AN} &= \frac{(480 \angle 0^\circ)(1 \angle -30^\circ)}{\sqrt{3}} \\ &= 277.13 \angle -30^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_{AN} is, $277.13 \angle -30^\circ$ V.

Step 3 of 12

Calculate the value of the voltage V_{BN} .

$$V_{BN} = \frac{V_{BC} \angle -150^\circ}{\sqrt{3}}$$

Substitute $480 \angle -120^\circ$ V for V_{BC} .

$$\begin{aligned} V_{BN} &= \frac{(480 \angle -120^\circ)(1 \angle -150^\circ)}{\sqrt{3}} \\ &= 277.13 \angle 90^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_{BN} is, $277.13 \angle 90^\circ$ V.

Step 4 of 12

Calculate the value of the voltage V_{CN} .

$$V_{CN} = \frac{V_{CA} \angle 90^\circ}{\sqrt{3}}$$

Substitute $480 \angle 120^\circ$ V for V_{CA} .

$$\begin{aligned} V_{CN} &= \frac{(480 \angle 120^\circ)(1 \angle 90^\circ)}{\sqrt{3}} \\ &= 277.13 \angle 210^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_{CN} is, $277.13 \angle 210^\circ$ V.

Step 5 of 12

(b)

Calculate the value of the current I_1 .

$$I_1 = \frac{V_{AN}}{48}$$

Substitute $277.13 \angle -30^\circ$ V for V_{AN} .

$$\begin{aligned} I_1 &= \frac{277.13 \angle -30^\circ}{48} \\ &= 5.77 \angle -30^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_1 is, $5.77 \angle -30^\circ$ A.

Step 6 of 12

Calculate the value of the current I_2 .

$$I_2 = \frac{V_{BN}}{40}$$

Substitute $277.13 \angle 90^\circ$ V for V_{BN} .

$$\begin{aligned} I_2 &= \frac{277.13 \angle 90^\circ}{40} \\ &= 6.93 \angle 90^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_2 is, $6.93 \angle 90^\circ$ A.

Step 7 of 12

Calculate the value of the current I_3 .

$$I_3 = \frac{V_{CN}}{60}$$

Substitute $277.13 \angle 210^\circ$ V for V_{CN} .

$$\begin{aligned} I_3 &= \frac{277.13 \angle 210^\circ}{60} \\ &= 4.62 \angle 210^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_3 is, $4.62 \angle 210^\circ$ A.

Step 8 of 12

Calculate the value of the current I_n .

$$I_n = -(I_1 + I_2 + I_3)$$

Substitute $5.77\angle -30^\circ$ A for I_1 , $6.93\angle 90^\circ$ A for I_2 and $4.62\angle 210^\circ$ A for I_3 .

$$\begin{aligned} I_n &= -(5.77\angle -30^\circ + 6.93\angle 90^\circ + 4.62\angle 210^\circ) \\ &= -(2\angle 60.14^\circ) \\ &= 2\angle -119.86^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_n is, $2\angle -119.86^\circ$ A.

Step 9 of 12

(c)

Calculate the reading of the wattmeter W1.

$$P_1 = (|I_1|)^2 (48)$$

Substitute 5.77 A for $|I_1|$.

$$\begin{aligned} P_1 &= (5.77)^2 (48) \\ &= (33.29)(48) \\ &= 1597.92 \text{ W} \end{aligned}$$

Therefore, the reading of the wattmeter W1 is, 1597.92 W.

Step 10 of 12

Calculate the reading of the wattmeter W2.

$$P_2 = (|I_2|)^2 (40)$$

Substitute 6.93 A for $|I_2|$.

$$\begin{aligned} P_2 &= (6.93)^2 (40) \\ &= (48.02)(40) \\ &= 1920.8 \text{ W} \end{aligned}$$

Therefore, the reading of the wattmeter W2 is, 1920.8 W.

Step 11 of 12

Calculate the reading of the wattmeter W3.

$$P_3 = (|I_3|)^2 (60)$$

Substitute 4.62 A for $|I_3|$.

$$\begin{aligned} P_3 &= (4.62)^2 (60) \\ &= (21.34)(60) \\ &= 1280.4 \text{ W} \end{aligned}$$

Therefore, the reading of the wattmeter W3 is, 1280.4 W.

Step 12 of 12

(d)

Calculate the value of the total power absorbed by the load.

$$P_T = P_1 + P_2 + P_3$$

Substitute 1280.4 W for P_3 , 1920.8 W for P_2 and 1597.92 W for P_1 .

$$\begin{aligned} P_T &= 1597.92 + 1920.8 + 1280.4 \\ &= 4799.12 \text{ W} \end{aligned}$$

Therefore, the value of the total power P_T absorbed by the load is, 4799.12 W.